

Detecting Shapes

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1 Introduction

Data originating from the real world often contain rich geometric and structural information, but this information is difficult to exploit using classical analysis methods. This is particularly true when the studied objects are represented as point clouds in high-dimensional spaces, such as in computer vision, biology, or scientific data analysis. Recognizing and comparing shapes from such representations therefore constitutes a non-trivial problem.

In this project, we focus on the recognition and automatic clustering of geometric shapes from three-dimensional point clouds. The considered dataset is composed of various shapes such as line segments, circles, disks, spheres, ellipsoids, as well as more irregular shapes. Although some of these shapes are geometrically distinct, they may share similar topological properties, which limits the effectiveness of purely geometric approaches.

To address this problem, we propose a method based on persistent homology, a central tool in computational topology. This approach makes it possible to capture global properties of shapes, such as connectivity, the presence of loops, or cavities, across multiple scales.

Our approach relies on the construction of simplicial filtrations from point clouds, followed by the computation of persistence diagrams. These diagrams provide a topological signature for each shape, which can then be compared using appropriate distance measures in order to quantify their similarity. This methodology allows for the automatic grouping of similar shapes while remaining robust to geometric transformations such as translations, rotations, or moderate deformations.

The remainder of this report presents the complete analysis pipeline, from data preprocessing to the comparison of topological signatures, before discussing the obtained results and the limitations of the proposed approach.

2 Method

Our approach follows a pipeline whose objective is to transform point clouds into comparable topological signatures. The overall process is divided into three main steps: data preprocessing, the construction of simplicial filtrations, and

the computation of persistent homology. These steps make it possible to extract robust topological invariants, which are then used to compare and cluster the studied shapes.

2.1 Data generation and preprocessing

We consider point clouds representing different geometric shapes embedded in R^3 . In order to develop and calibrate the method in a controlled setting, we generate a synthetic dataset from ideal geometric models, to which low-amplitude Gaussian noise is added. Random rotations and translations are also applied to ensure that the method is invariant with respect to the position and orientation of the shapes.

Before any topological computation, each point cloud is centered and normalized. Centering ensures translation invariance, while normalization makes it possible to compare shapes of different sizes using consistent distance parameters. This preprocessing step is essential for the construction of the filtrations described below.

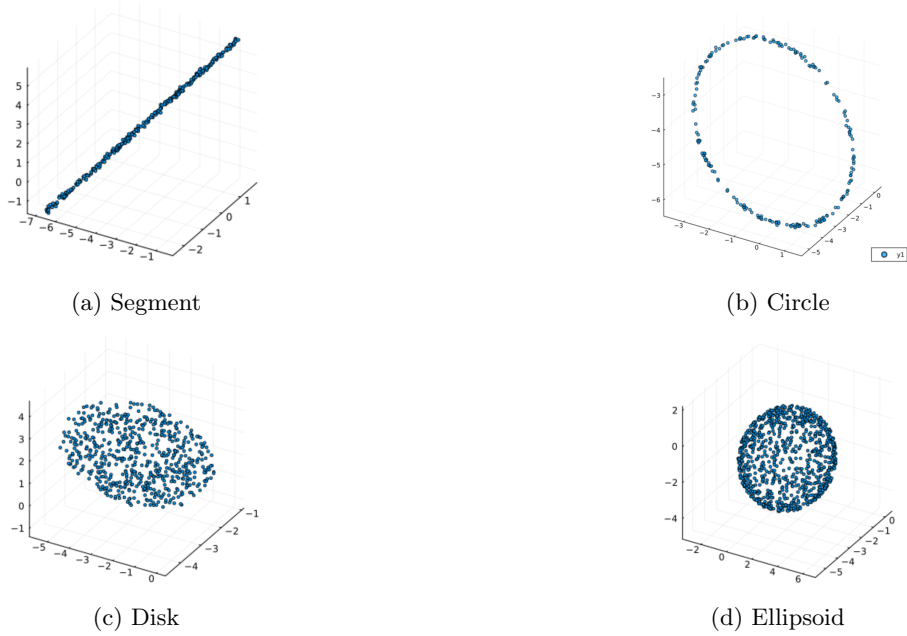


Figure 1: Point clouds representing different generated geometric shapes

2.2 Filtrations and simplicial complexes

From the preprocessed point clouds, we construct filtrations of simplicial complexes in order to approximate the underlying topological structure of the data.

Initially, we implemented functions for two classical types of filtrations: Vietoris–Rips complexes and Čech complexes, which were built manually in order to better understand their behavior and theoretical differences.

However, these constructions quickly proved to be computationally expensive, especially when the number of points increases or when several homology dimensions are considered. For this reason, we subsequently chose to focus exclusively on Vietoris–Rips filtrations, which provide a good trade-off between topological expressiveness and computational complexity.

The final computations are performed using the specialized library Ripserer, which enables efficient and robust computation of persistent homology directly from point clouds, while ensuring reproducibility and stability of the results.

Distances between persistence diagrams are combined in order to obtain a global dissimilarity matrix that integrates information from multiple homology dimensions. Initially, this matrix is used to perform agglomerative hierarchical clustering. This step makes it possible to explore the structure of the data and to verify that topologically similar shapes are correctly grouped together, while also highlighting certain ambiguous cases.

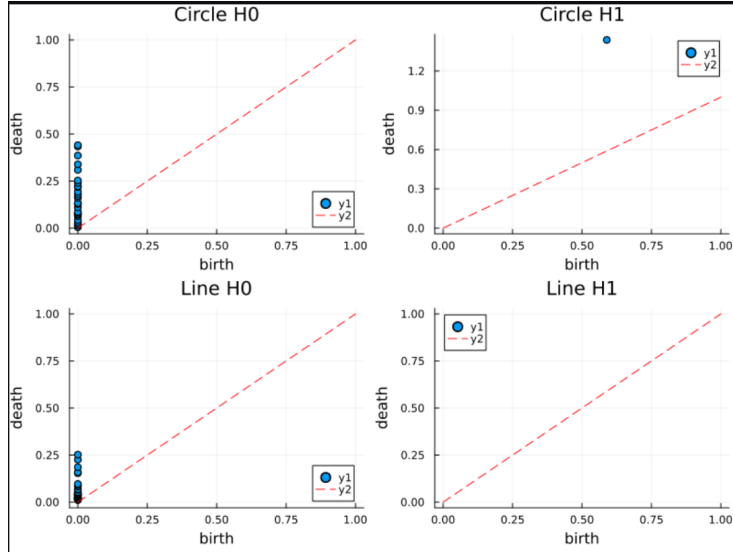


Figure 2: Comparison of the homology groups H_0 and H_1 for a circle and a segment. Both shapes are connected (identical H_0), but only the circle exhibits a non-trivial class in dimension 1.

To compare topological signatures, we measure the dissimilarity between persistence diagrams using the Wasserstein distance. This comparison is carried out separately for each homology dimension (H_0 , H_1 , H_2), resulting in a distance matrix $D^{(k)}$ for each dimension. We then construct a global distance

by combining these matrices in the form of a weighted sum

$$D = \sum_k w_k D^{(k)},$$

where the coefficients w_k allow us to adjust the relative importance of each dimension.

Before performing supervised classification, we apply agglomerative hierarchical clustering to the matrix D in order to explore the structure of the data and to verify that shapes that are close from a topological point of view are effectively grouped together. Once this consistency is observed, we use the same matrix D within a k -nearest neighbors (k-NN) classifier. Performance is finally evaluated using leave-one-out cross-validation: each point cloud is removed in turn and then predicted based on its k nearest neighbors in the distance matrix.

3 Results: simple geometric shapes

In a first stage, we restrict the study to a limited set of simple geometric shapes. The experiments focus on four shapes: line segments, circles, disks, and ellipsoids. For each type of shape, ten independent point clouds are generated, centered at the origin and normalized so that the farthest point lies at unit distance.

3.1 Choice of homology dimension

For this set of shapes, we observed that persistent homology in dimension H_0 is the most informative for clustering. Although all shapes are connected at the end of the filtration, they exhibit very different behaviors in terms of point density and spatial distribution, which translates into distinct merging dynamics of connected components.

In contrast, homology in dimension H_1 provides only limited information in this setting. With the exception of circles, disks and ellipsoids do not exhibit persistent cycles, which makes H_1 poorly discriminative for distinguishing these shapes. The results presented in the following therefore rely primarily on the analysis of H_0 .

3.2 Influence of the number of points

We study the influence of the number of sampled points on clustering quality. For a small number of points (30 and 50), hierarchical clustering based on H_0 correctly distinguishes segments and circles, but confusions appear between disks and ellipsoids. When the number of points is increased to 80, the four shapes are correctly separated.

These results indicate that the method becomes more reliable when the sampling is sufficiently dense, particularly for topologically similar shapes.

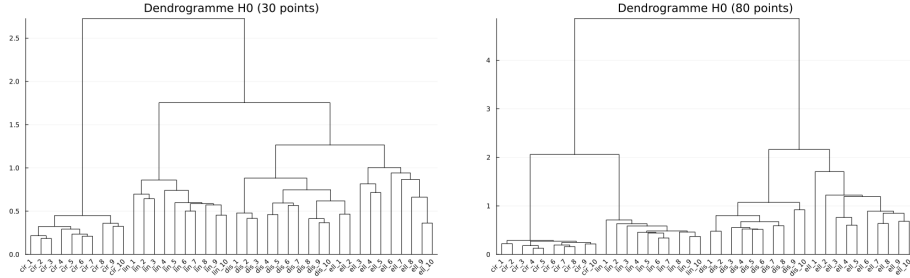


Figure 3: Dendrograms obtained by hierarchical clustering based on topological distances H_0 , for different numbers of points. On the left, a sparse sampling (30 points) leads to a partial separation of the shapes. On the right, increasing the number of points (80 points) allows for a better discrimination of the four considered shapes.

3.3 Influence of noise

We then analyzed the impact of noise by fixing the number of points to 30 and progressively reducing the variance. In this setting, circles and segments remain correctly distinguished, but confusions between disks and ellipsoids persist, even for very low noise levels.

This behavior suggests that these confusions are not primarily due to noise, but rather to the insufficient number of points and to the fact that disks and ellipsoids share the same homology. The information captured by H_0 therefore remains insufficient to reliably separate them in this regime.

3.4 Modification of ellipsoid geometry

In order to reduce the similarity between disks and ellipsoids, we modified the geometry of the latter by increasing their thickness. More precisely, the parameter c was increased from 1.5 to 2.5, while the parameters $a = 4.0$ and $b = 2.5$ were kept unchanged. This modification makes the ellipsoids less similar to disk-like shapes and improves their separation in the space of topological signatures.

In this case, even with only 30 points, clustering based on H_0 correctly separates all shapes. This result shows that reliable clustering can be achieved with few points when shapes are sufficiently distinct, whereas denser sampling is required to distinguish topologically equivalent shapes.

3.5 Classification using k-nearest neighbors

After validating the consistency of the topological distances through clustering, we use the same distance matrix in a supervised setting, using a k-nearest neighbors classifier. Performance is evaluated using leave-one-out cross-validation.



Figure 4: Comparison between two ellipsoid geometries. On the left, a thin ellipsoid close to a disk-like structure; on the right, a thicker ellipsoid. Increasing the thickness modifies the effective sampling of the shape and improves topological separation, even with a limited number of points.

With the initial noise level and a sampling limited to 30 points per point cloud, the method achieves an overall accuracy of

$$accuracy = 0.95,$$

with only two classification errors. These errors correspond to confusions between topologically equivalent shapes, in particular between disks and ellipsoids, which is consistent with previous observations and with the theoretical limitations of persistent homology in this setting.

When the number of points is increased, or when the shapes are made geometrically more distinct despite having similar homology, these confusions disappear and the classification becomes perfect.

4 Results: extended set of shapes

After validating the method on simple shapes, we extend the study to a richer set of geometric shapes in order to evaluate the robustness and generalization capability of the approach. The considered shapes are the following: circle, line segment, disk, ellipsoid, irregular *potato*-like shape, sphere, and torus. For each shape, eight point clouds are generated, each consisting of 70 points.

In contrast to the case of simple shapes, the analysis here relies on a combination of several homology dimensions, which makes it possible to capture more diverse topological structures, in particular the presence of cycles and cavities.

4.1 Classification using k-nearest neighbors

Several weightings of the topological distances are tested within the k-nearest neighbors classifier. The best results are obtained for the weights (0.5, 3.0, 0.5).

In this setting, the method achieves an overall accuracy of

$$accuracy = 0.964,$$

which corresponds to only two classification errors over the entire dataset.

4.2 Error analysis

The two observed errors are summarized in Table 1.

Index	True label	Predicted label
29	ellipsoid	torus
44	sphere	ellipsoid

Table 1: Classification errors observed during leave-one-out validation.

The first error corresponds to an ellipsoid classified as a torus. This confusion can be explained by a particular geometry of the sampled shape, combined with noise and discrete sampling, which may generate artificial cycles in the filtration and bring its topological signature closer to that of a torus.

The second error concerns a sphere classified as an ellipsoid. These two shapes are topologically equivalent, sharing the same homology, and differ only in their geometry. This confusion is therefore theoretically consistent and illustrates a natural limitation of purely topological descriptors.

4.3 Discussion

These results show that the method is able to effectively distinguish a wide variety of shapes, including complex ones such as the torus or the *potato*-like shape. The observed errors are rare and can be explained by intrinsic topological similarities or by effects related to sampling and noise.

Overall, persistent homology, combined with distance-based comparison and a simple k-nearest neighbors classifier, constitutes a robust and relevant tool for the automatic recognition of shapes from point clouds.

5 Persistence Silhouette

(this section is added by Eske)

In addition to the direct shape classification using the Wasserstein distance on persistence diagrams, we also investigated an approach based on classifying persistence diagrams via vectorized representations. For this purpose, we followed the definitions introduced in “Stochastic Convergence of Persistence Landscapes and Silhouettes” by F. Chazal et al.

Starting from the persistence diagrams, we computed the corresponding persistence silhouettes. The silhouettes of different homological dimensions were then concatenated to form a single feature representation. During this concatenation step, we also experimented with assigning different weights to the individual dimensions in order to account for their varying importance.

On the resulting sequences, we applied several clustering algorithms, namely k-means, hierarchical clustering, and DBSCAN. Among these methods, k-means

clustering yielded the best results. Nevertheless, despite this improvement, it was not possible to correctly classify all shapes using this approach. While sometimes the results were also perfect.

Unfortunately the code as it is right now, need an idea of how many clusters we want to have in the end. This is an issue if there is an unknown number of shapes contained in the dataset.

When we tried to test the cde on the given data set, we had an issue with Ripserer, which due to a time issue, we were not able to solve before submitting the report.

6 Conclusion

In this project, we investigated the use of persistent homology for the automatic recognition of geometric shapes from three-dimensional point clouds. By constructing Vietoris–Rips filtrations and comparing persistence diagrams using appropriate distances, we were able to extract robust topological signatures and exploit them in both unsupervised and supervised settings.

Experiments conducted on a restricted set of simple shapes showed that the choice of homology dimension is crucial. In particular, homology in dimension H_0 proved to be the most informative for distinguishing segments, circles, disks, and ellipsoids, by capturing differences in connectivity and point density within the point clouds. The confusions observed between certain shapes were interpreted as natural limitations related to their topological equivalence and to insufficient sampling.

Extending the method to a richer set of shapes confirmed its generalization capability. By combining several homology dimensions and using a k-nearest neighbors classifier, we achieved high performance, with a classification rate above 96%. The rare observed errors are topologically consistent and illustrate the limitations of purely topological descriptors for distinguishing geometrically different but topologically equivalent shapes.

These results demonstrate that persistent homology is a relevant tool for the analysis and recognition of shapes from point clouds. Possible directions for improvement include the use of alternative vector representations of persistence diagrams, the combination with geometric descriptors, and the study of more complex or higher-dimensional datasets.

7 Devision of work

Producing the pointclouds, Programming the knn part, Report - Eliot
Programming the filtrations, persistence diagrams with ripserer, silhouette, distance matrices, plots, clustering, Testing the test data with silhouette clustering, silhouette part in report- Eske